Table 1: Quantifications of Two Variables

Statement	When True?	When False?
$\forall x \forall y P(x,y)$ $\forall y \forall x P(x,y)$	$P(x,y)$ is true for every pair (x,y)	There is a pair x,y for which $P(x,y)$ is False
$\forall x \exists y P(x,y)$	For every x there is a y for which $P(x,y)$ is True	There is an x such that $P(x,y)$ is False for every y
$\exists x \forall y P(x,y)$	There is an x for which $P(x,y)$ is true for every y	For every x there is a y for which $P(x,y)$ is False
$\exists x \exists y P(x,y)$ $\exists y \exists x P(x,y)$	There is a pair (x,y) for which $P(x,y)$ is True	$P(x,y)$ is False for every pair (x,y)

Example 5: Let $Q(x,y,z)$ be the statement "$x+y=z$". What are the truth values of the statements $\forall x \forall y \exists z\, Q(x,y,z)$ and $\exists z \forall x \forall y\, Q(x,y,z)$, where the domain of all variables consists of all real numbers?

Soln: Suppose that x and y are assigned values. Then, there exists a real number z such that $x+y=z$. Consequently, the quantification $\forall x \forall y \exists z\, Q(x,y,z)$ which is the statement "For all real numbers x and for all real numbers y there is a real number z such that $x+y=z$," is true.

- The order of the quantification here is important, because the quantification $\exists z \forall x \forall y\, Q(x,y,z)$ which is the statement "There is a real number z such that for all real numbers x and for all real numbers y it is true that $x+y=z$," is false, because there is no value of z that satisfies the equation $x+y=z$ for all values of x and y.

1.5.4 Translating Mathematical Statements to Statements Involving Nested Quantifiers

Example 6: Translate the statement "The sum of two positive integers is always positive." into a logical expression.

Soln: $\forall x \forall y\, ((x>0) \land (y>0) \Rightarrow (x+y>0))$ where the domain of both x and y are the set of integers.